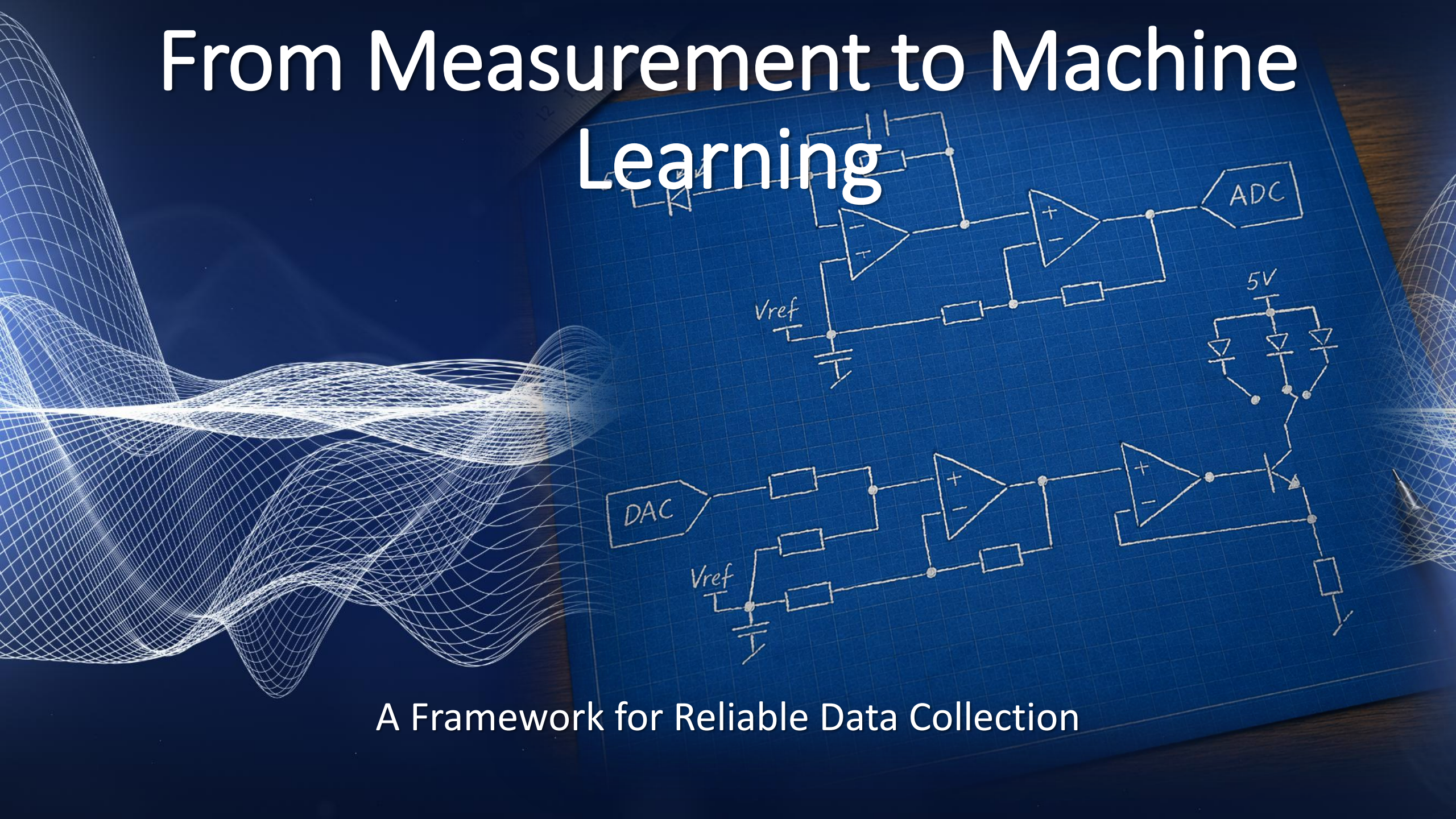
From Measurement to Machine Learning

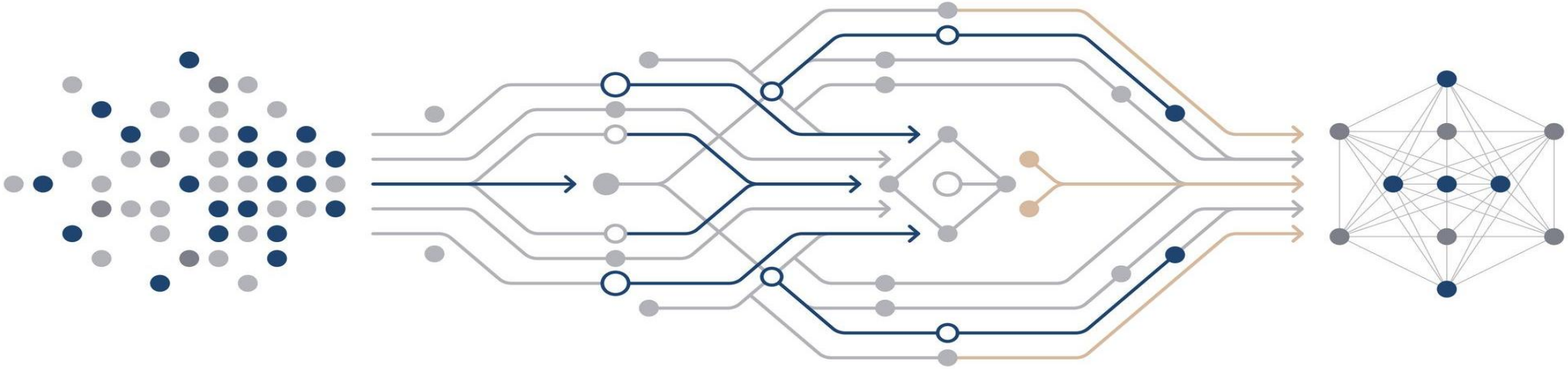


A Framework for Reliable Data Collection

The Age of Data

- Data-driven models
- Massive datasets
- “Quality data – Quality models”

- What is Quality data?



Quality Data

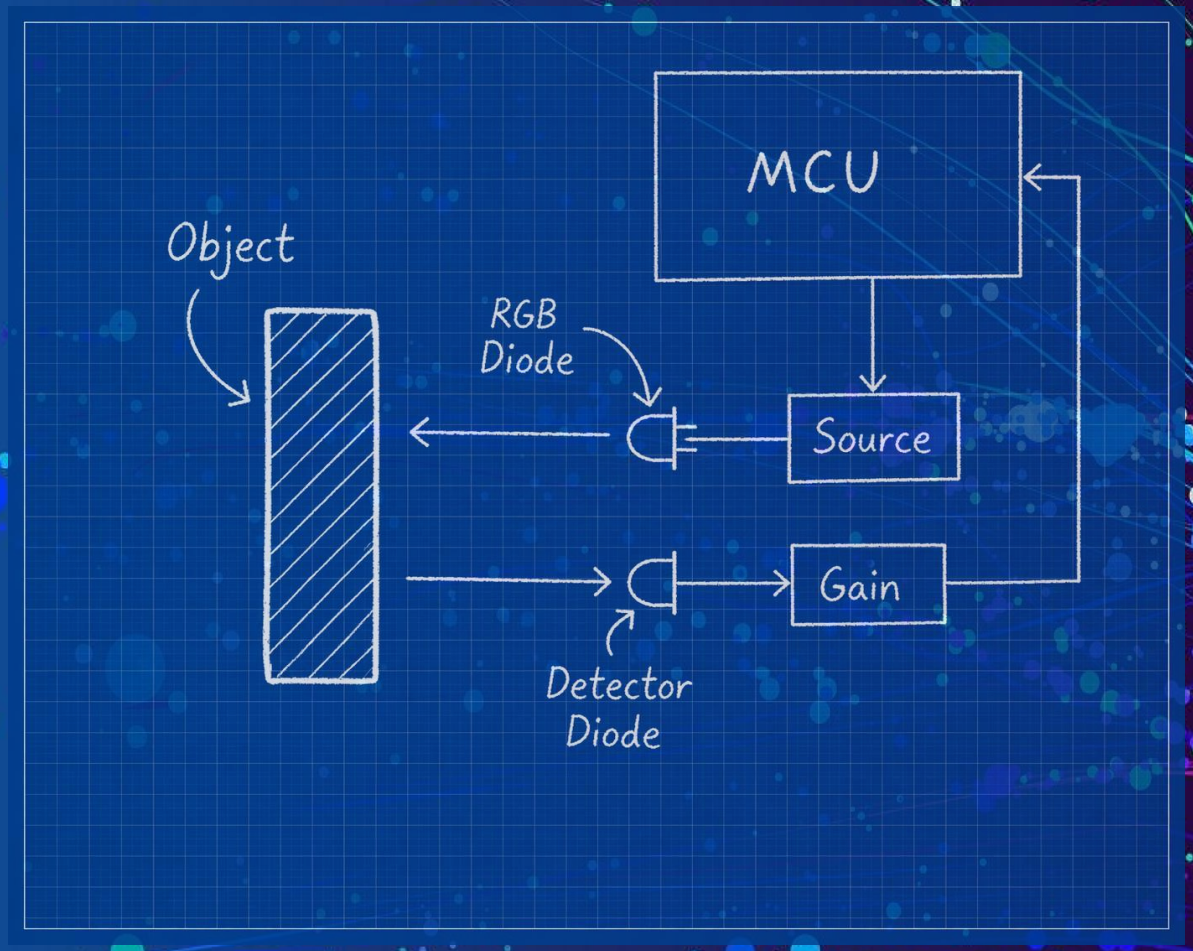
- Physical Fidelity
- Accurate Ground Truth
- Repeatability
- Noise
- Dynamic Range and Resolution

A Simple Setup

Infer Distance from RGB Data

- Simple measurement system
- RGB intensity input

- Can we uniquely determine distance?



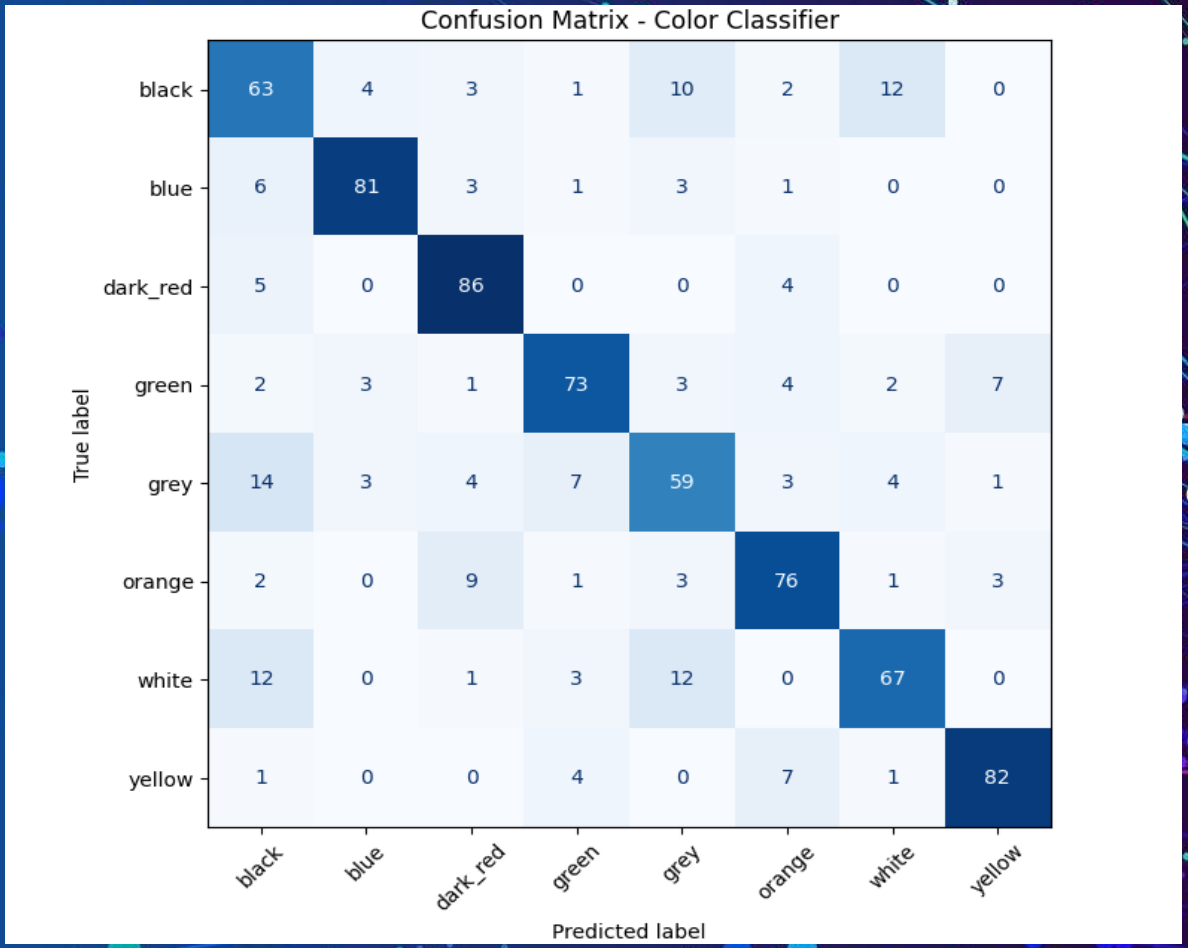
A Simple Setup

Infer Distance from RGB Data

- Simple measurement system
- RGB intensity input

- Can we uniquely determine distance?

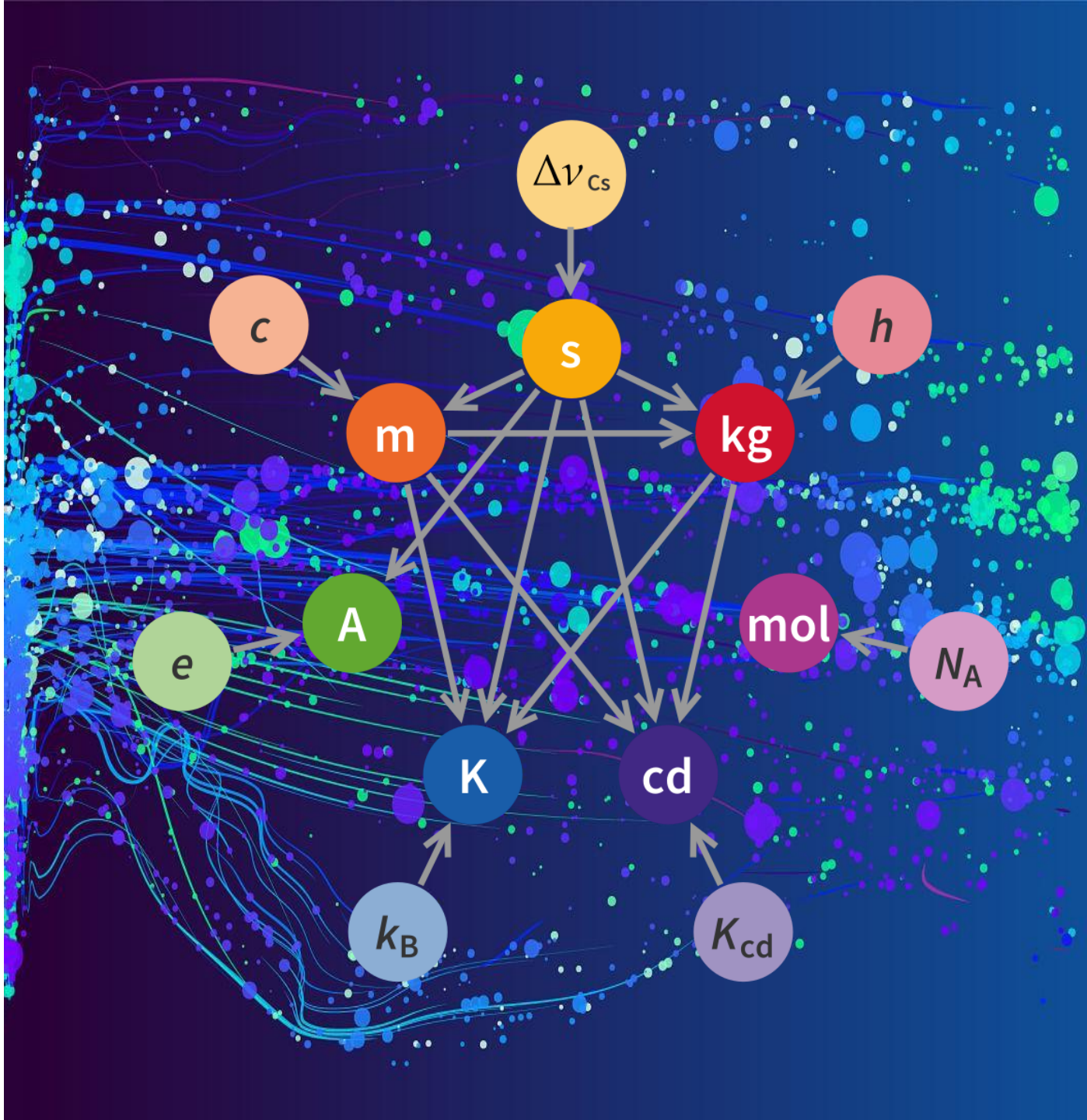
- **Is this surprising?**
  Black / White confusion?
- **Will more data fix this?**



Metrology

The science of Measurement

- Engineer the system for **clean observation**
- Quantify what we know
  -> And what we **don't know**
- Foundation of **High-quality data**
- Not a dataset problem
  -> A Physics problem

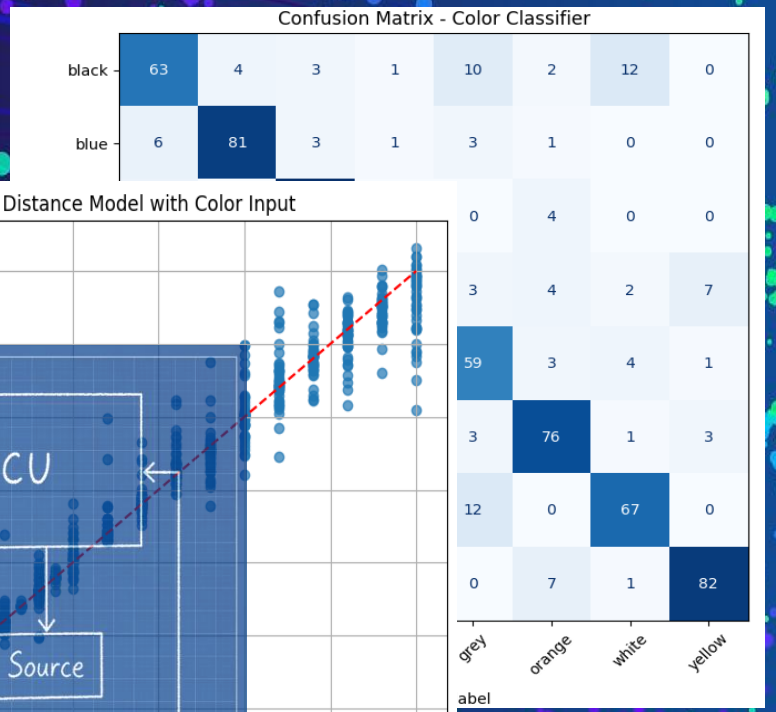


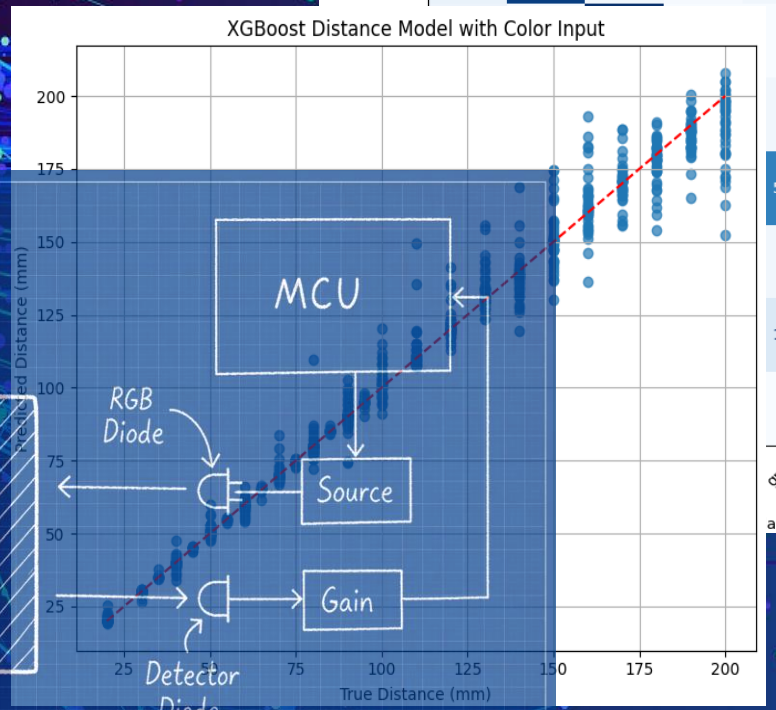
A Unifying Framework For Data Collection

- Define the **measurement objective**
- Engineer the **observation**
- Document **method and conditions**
- Quantify **uncertainty and confidence**
- Ensure **reproducibility and traceability**

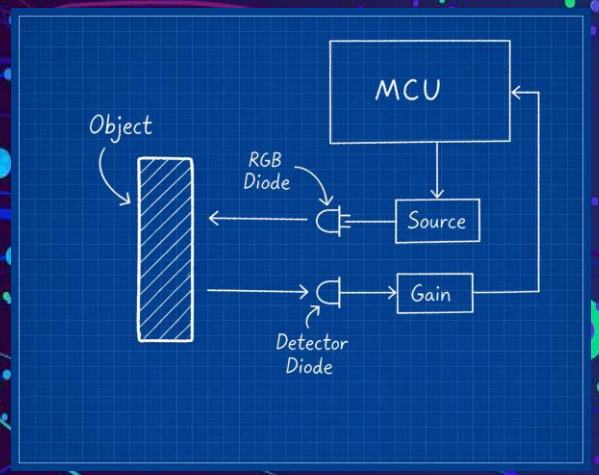
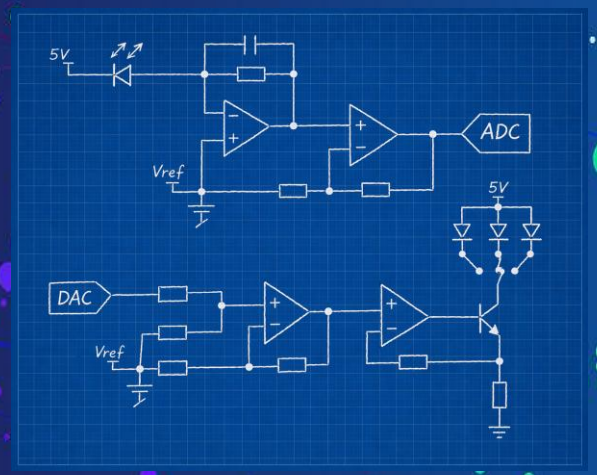
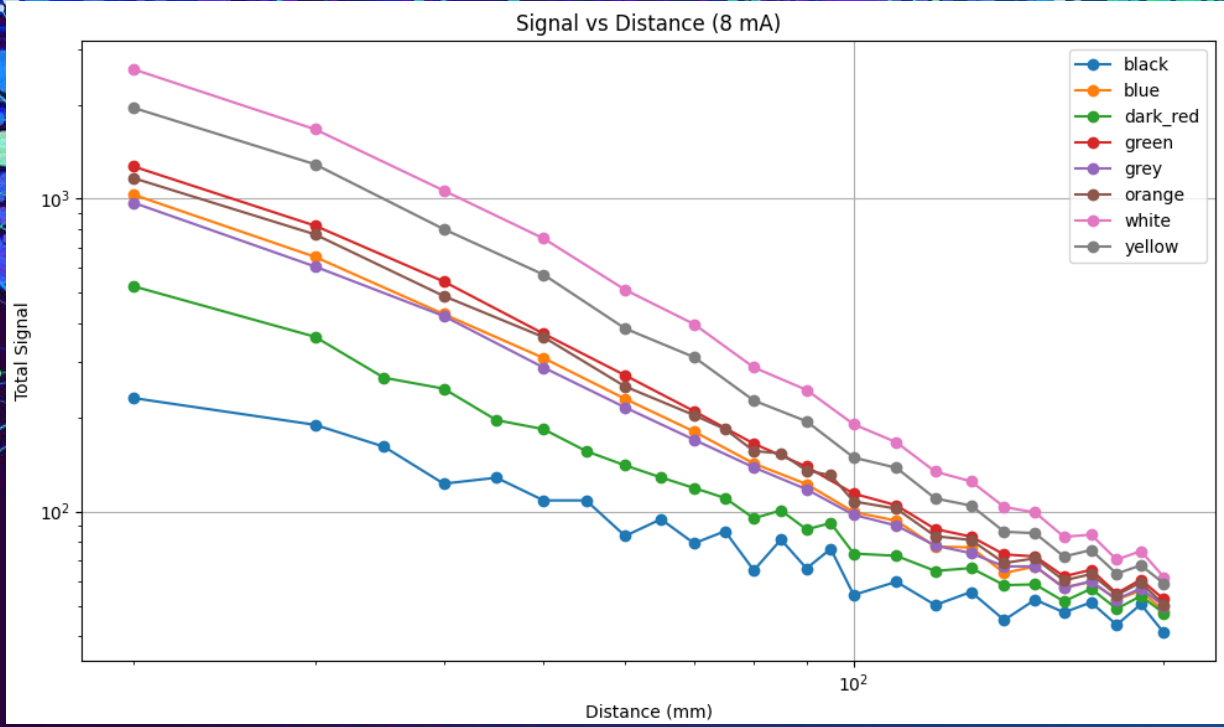
General Guide to Measurements
APMS Aalborg University

APMS Department





Framework

- The Example

- Infer distance from RGB intensity
- 2 stage model:
  -> Colour Classifier **->** Regression
- DSP pre-processing applied
- Simple, controlled
  measurement setup